

PRANJAY DHAWAN

Chandigarh

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My Portfolio

Education

Thapar Institute of Engineering and Technology

August 2023 – July 2026

Bachelor of Engineering in Electronics and Communication Engineering

Patiala, Punjab

Chandigarh College of Engineering and Technology

August 2020 – June 2023

Diploma in Engineering in Electronics and Communication Engineering

Chandigarh

Experience

Thapar Institute of Engineering and Technology

August 2026 – Present

Research Associate

Patiala, Punjab

- Working on the overall structure and system integration of a walking-assist exoskeleton.
- Implementing ROS 2 and Simulink in the project, feeding collected human data into the exoskeleton to assist users in walking.

Zipbolt Innovation Pvt Ltd

January 2026 – July 2026

Product R&D Engineer - Internship

Chandigarh

- Developed hardware architecture and PCB for the ZipSure AI EV monitoring system for real-time battery analytics consisting of sensory node and gateway node.
- Designed and prototyped EV-compatible enclosures for ZipSure AI using SolidWorks and in-house 3D printing.
- Enhanced firmware for ESP32-based IoT devices at charging station by adding advanced debugging and diagnostics features.

The MindCrafters

February 2022 – July 2023

Core Electronics Engineer - Internship

Chandigarh

- PCB Design and Electronics: Engineered and fabricated 4 custom PCBs using EasyEDA for robotic applications, ensuring seamless integration of electronic components.
- IoT and Robotics Projects: Spearheaded the development of 4 humanoid robot prototypes and 3 IoT systems utilizing ThingSpeak, improving project performance and data analytics capabilities.
- 3D Printing and CAD Design: Executed 3D printing and CAD design for 50+ complex models using SketchUp, contributing to projects development.

Projects

Drone - Garud | Altium, Pixhawk, Mission Planner, Solidworks, PrusaSlicer

May 2025 – February 2026

- Redesigned and optimized 4 structural points, reducing chassis weight from 650 g to 450 g while improving aerodynamic efficiency and mechanical robustness.
- Developed the electronic integration for a Pixhawk-based drone—covering GPS/IMU interfacing, telemetry links, and power routing—and executed extensive flight-testing in Mission Planner, optimizing PID tuning, thrust balance, vibration isolation, and GPS stability for reliable manual flight.

Martian Rover - Ganga | Altium, Teensy 4.1, Solidworks, PrusaSlicer

April 2025 – February 2026

- Engineered a robust system architecture with a simplified main PCB integrating dual Teensy 4.1 microcontrollers for full-rover control in IRC mission scenarios.
- Designed a custom Power Distribution Unit featuring dual 19V boost converters, dual bucks outputting 5V and 8V, 12V short-circuit protection, and a 5-channel solid-state relay to ensure reliable power delivery during field operations.
- Led the design, optimization, and validation of 25+ structural components throughout rover manufacturing, improving mechanical performance while adhering to the IRC weight limit of 65 kg.

VocalEyes | Python, Raspberry Pi 5, Solidworks, PrusaSlicer

February 2025 – December 2026

- Developed “VocalEyes,” a Raspberry Pi-based assistive device integrating I2S DAC Module and Pi Camera 3 Module with modular integration, achieving reliable real-time object awareness in diverse environments and providing low-cost accessibility to the visually impaired.
- Designed and 3D printed a self-ideated universal cap mount for Raspberry Pi 5 and other accessories, ensuring ergonomic usability and adaptability for users of any head size.

Martian Rover - Alaknanda | C/C++, Teensy 4.1, Solidworks, PrusaSlicer

August 2024 – February 2025

- Designed and implemented the electronics subsystem, integrating 6 sensors, 18 motor drivers, and power management circuits to ensure stable operation in diverse terrains.
- Led and mentored team in design and 3D printing of 45+ structural components using PrusaSlicer, significantly optimizing the rover’s mechanical stability and reducing overall weight.

DIY Water Meter | EasyEDA, Arduino IDE, ESP32, C++

June 2022 – September 2022

- Created a system to accurately measure water parameters, achieving consistent readings across 5 environmental conditions.
- Integrated advanced API functionalities into IoT systems to facilitate remote readings from the DIY water meter project, achieving precise measurements across 5 different parameters.

Publications in International and National Conferences

International Conf. on Intelligent Robots and Systems (IEEE IROS 2026) | *Accepted - Poster* September 2026

- Co-authored “Alaknanda: A ROS 2-Enabled Modular Mars Rover for Field and Astrobiological Exploration,” accepted for poster presentation at the Robotic One-Shot Exploration from Deep Ocean to Deep Space (IEEE IROS, 2026 - ROSE) Workshop, Pittsburgh, USA; contributed electrical architecture design and implementation along with power system design and validation, under the mentorship of Dr. Ashish Singla and Dr. Sachin Kansal.

International Conference on Smart Cyber Physical Systems (ICSCPS, 2026) | *Published* January 2026

- Contributed to the research paper by designing the system architecture of an ESP32-based outdoor multi-sensor tracker, integrating GNSS, thermal and RGB sensing, power management, and IoT data transmission into a unified embedded platform under the guidance of Dr. Amanpreet Kaur.

7th International Conference of The Robotics Society (AIR, 2025) | *Published* June 2025

- Contributed to research on 4 custom PCB design, thermal management, and battery modeling for a multi-subsystem Martian rover, presented at a leading robotics conference under the mentorship of Dr. Ashish Singla and Dr. Sachin Kansal.

3rd IFToMM for SDG Conference, 2025 | *Extended Abstract* June 2025

- Presented an extended abstract on a custom Martian-rover power system achieving 18 Nm torque and 86.7% efficiency while reducing heat-induced losses, guided by Dr. Ashish Singla and Dr. Sachin Kansal.

Leadership/Extracurricular

Overall Student Coordinator July 2025 – July 2026

Mechatronics and Robotics Society, Thapar Institute of Engineering and Technology Patiala, Punjab

- Directed 2 teams comprising 50+ students under the guidance of Dr. Ashish Singla (President) and Dr. Sachin Kansal (Vice-President) for the International Rover Challenge, overseeing electronics integration, manufacturing, and 3D printing tasks, and led a separate team for the International Space Drone Challenge, managing overall drone development.
- Supervised the complete development of 2 ELC Projects integrating 50 electronic components for MARVIN - Cleaning Bot and Reconfig - Reconfigurable Bot, ensuring efficient system design and testing.
- Contributed to the design and integration of all systems in 3 minor projects under the MARS Society, improving prototyping efficiency and overall system reliability.

Lead - Electronics Department June 2024 – June 2025

Mechatronics and Robotics Society, Thapar Institute of Engineering and Technology Patiala, Punjab

- Directed a team of 30+ students under the guidance of Dr. Ashish Singla (President) and Dr. Sachin Kansal (Vice-President) for the International Rover Challenge, managing electronics and 3D printing tasks.
- Supervised the electronics development of an ELC Project integrating 42 electronic components for a surveillance rover designed for military applications.
- Contributed to the design and integration of electronics systems in 5 ongoing minor projects under the MARS Society, enhancing prototyping efficiency and system reliability.

Technical Skills

Languages: C/C++, Python, HTML.

Frameworks/Tools: Altium, EasyEDA, AutoDesk Eagle, Solidworks, Arduino IDE, STM32Cube IDE, Github, Vercel, MATLAB and ThingSpeak by MathWorks, VSCode, SketchUp, PrusaSlicer, Xilinx Vivado, Mentor Graphics.

Core Skills Knowledge: Schematic Capture, Power Electronics, PCB Design, Analog Electronics, Microcontrollers, Embedded Hardware, 3D Designing and Printing, Computer Architecture, Digital System Design.

Certifications

- Certificate of Participation at International Conference on Smart Cyber Physical Systems, 2026
- Certificate of Completion of Qualcomm Technologies' 5G Introductory Level Course
- Certificate of Completion of 21st Century Technologies Course
- Certificate of Completion of The Altium Education PCB Basic Design Course
- Certificate of Participation at Advances In Robotics (AIR) 2025 7th International Conference of The Robotics Society
- Certificate of Participation in the Workshop on "Dynamics of Robotics System"
- Certificate of Participation for HackOwasp 6.0 Hackathon.
- Certificate of Appreciation for successful completion of ELC activity of IOT BASED SYSTEMS.
- Certificate of Appreciation for successful completion of ELC activity of DC Power Supply.
- Certificate of Appreciation for Microsoft Entrepreneurship Competition.

Honors and Awards

- Awarded for Champion at International Rover Competition, 2026.
- Awarded for 1st Runner Up at International Space Drone Competition, 2026.
- Awarded for 1st Runner Up at International Rover Competition, 2025.
- Awarded a 1 lakh Academic Scholarship from Thapar Institute of Engineering and Technology.